Truth Vertex Distribution for MC Events With and Without a Reconstructed Vertex

photon10 / jet12 PYTHIA8 samples, single- and double-interaction production

PPG12 Analysis Team (automated report)

April 13, 2026

Motivation

In `RecoEffCalculator_TTreeReader.C` (efficiency pipeline) and `ShowerShapeCheck.C`, MC events whose reconstructed vertex falls outside the range of the data/MC `h_vertexz` histogram ($[-100, +100]$ cm) are assigned `vertex_weight = 0` and therefore drop out of both reco and truth histograms. The overwhelming majority of these "outside-range" events are the sentinel `vertexz` $= -9999$ cm written by `CaloAna24.cc` when the MBD vertex finder produces no `GlobalVertex::MBD` vertex.

The efficiency calculation is therefore conditional on the existence of an MBD reco vertex in both data and MC. This is self-consistent only if the MC correctly models the MBD vertex-finding efficiency. In particular, if the truth vertex distribution of "has-reco-vertex" events differs systematically from that of "no-reco-vertex" events, then the set of events actually used in the measurement is a biased subsample of the fiducial truth population.

This report compares the *truth* vertex-z distribution between events with and without a successfully reconstructed vertex, for four samples: photon10 and jet12, each in their nominal (single-interaction) and double-interaction productions.

Setup

- Samples (`slimtree` from `CaloAna24`):
  - `anatreemaker/macro_maketree/sim/run28/photon10/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/photon10_double/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/jet12/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/jet12_double/condorout/combined.root`
- "Has reco vertex" $\Leftrightarrow$ `vertexz > -999`.
- Histograms: 80 bins over `vertexz_truth` $\in [-200, +200]$ cm, unweighted event counts (no cross-section reweighting applied).
- Macro: `efficiencytool/TruthVertexRecoCheck.C` (fill) and `efficiencytool/TruthVertexRecoCheckDra` (plot).

Event counts

Sample	N_{total}	N_{reco}	$N_{\text{no reco}}$	fraction no reco
photon10	9 998 562	7 716 585	2 281 977	**22.82 %**
photon10_double	9 997 075	9 491 223	505 852	**5.06 %**
jet12	10 000 000	7 689 530	2 310 470	**23.10 %**
jet12_double	9 997 000	9 485 384	511 616	**5.12 %**

Table 1: Fraction of events without an MBD reco vertex. The double-interaction productions have $\sim 4.5\times$ higher MBD vertex-finding efficiency than their single-interaction counterparts.

Shape comparison

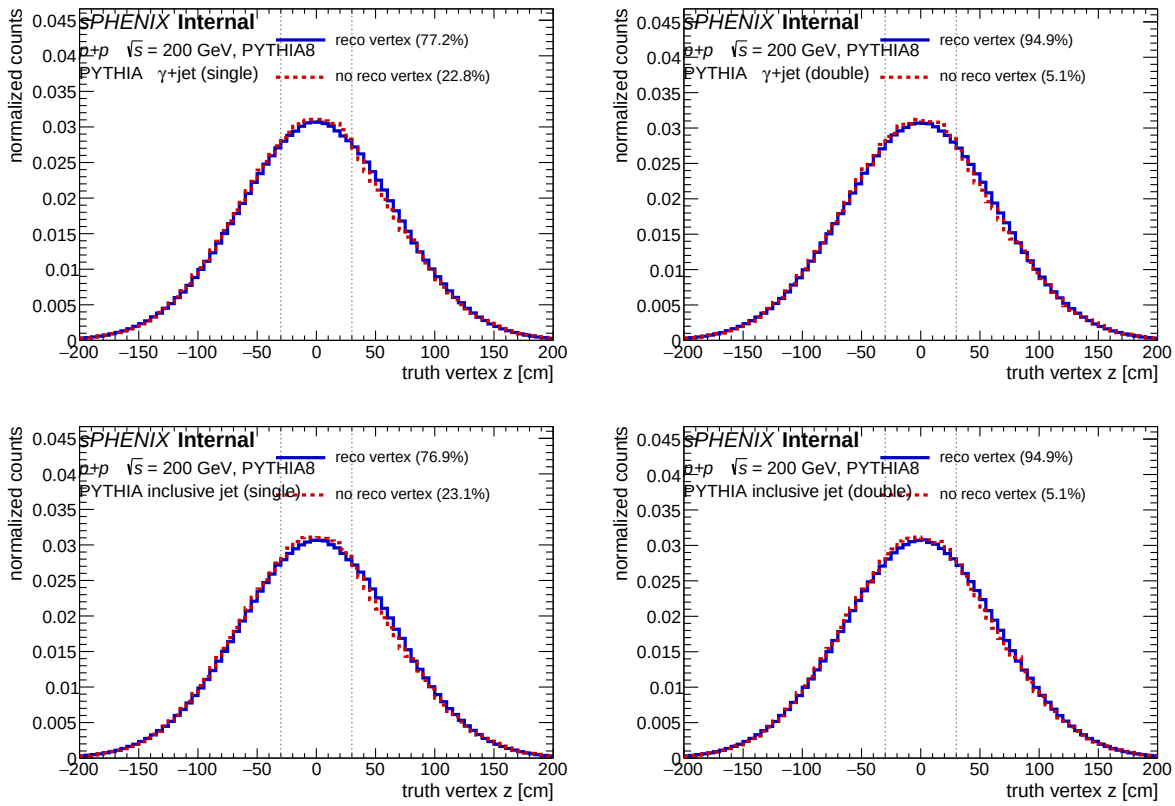

Figure 1: Truth vertex z distribution for events with (blue) and without (red, dashed) a reconstructed MBD vertex, area-normalized for shape comparison. Columns: single interaction (left) and double interaction (right). Rows: photon10 (top) and jet12 (bottom). Dotted vertical lines: analysis cut at $|z_{\text{vtx}}| = 30$ cm. Numbers in the legend: fraction of the total sample.

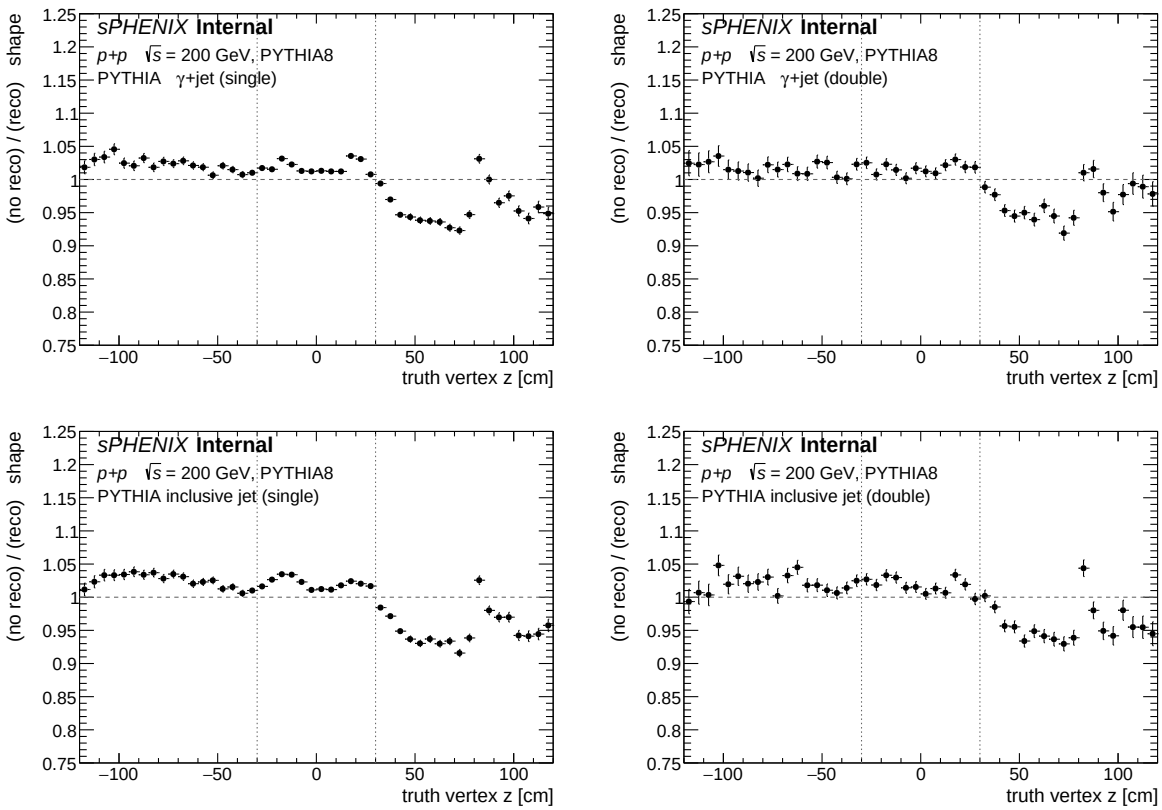

Figure 2: Ratio (no-reco-vertex shape) / (reco-vertex shape) as a function of truth vertex z, same samples as above. Y-axis zoomed to $[0.75, 1.25]$. Dotted vertical lines: $|z_{\mathrm{vtx}}| = 30$ cm analysis window.

Findings

1. **Global shapes are very similar.** In all four samples the truth vertex distributions of with- and without-reco events are both centered at $z = 0$ with the ~ 75 cm wide beam spot, and overlap at the few-percent level inside the analysis window.

2. **Double-interaction productions have dramatically higher MBD vertex efficiency** (Table 1). The fraction of events with no reco vertex drops from $\sim 23\%$ (single) to $\sim 5\%$ (double). This is physically expected: a second concurrent collision provides additional MBD hits, making the vertex finder more likely to succeed. The photon10 and jet12 samples agree on this fraction to within 0.3 percentage points, indicating the effect is driven by the event pile-up level rather than the physics process.

3. **Small, consistent z-asymmetry in the shape ratio.** The (no-reco / reco) shape ratio is not flat at unity: it is systematically ~ 1.02–1.05 for $z < 0$ and ~ 0.92–0.97 for $z > 0$, with the strongest deviation around $z \simeq +50$ to $+80$ cm where the ratio dips to ~ 0.90. The same asymmetric pattern appears in all four samples and is therefore a detector effect rather than a physics-process effect. It indicates that the MBD vertex finder is slightly more efficient at positive z than at negative z.

4. **Inside the analysis window the bias is small.** For $|z| < 30$ cm the ratio stays within $\sim \pm 5\%$ of unity, and it is nearly symmetric about zero at the peak of the distribution. Integrated over the analysis window, the "has reco" subsample is $\lesssim 1\%$ biased in its mean truth z relative to the full sample, which is unlikely to move the cross-section through the $|z| < 30$ cm acceptance factor.

5. **Outside $|z| > 30$ cm, events without a reco vertex have a modestly narrower tail**

on the $+z$ side than those with one. The $\sim$10% dip at $z \simeq +50$–80 cm is where the two samples disagree most.

Implications for the vertex reweighting

The vertex reweighting at `RecoEffCalculator_TTreeReader.C:1331-1372` sets `vertex_weight = 0` for events with $|$`vertexz`$| > 100$ cm, which drops all $\sim$23% (single) / $\sim$5% (double) of events without a reco vertex. Within the $|z| < 30$ cm analysis window the truth vertex distribution of the surviving sub-sample is shown here to differ from that of the dropped sub-sample at the few-percent level (approximately symmetric about zero). The corresponding bias on the fiducial acceptance factor is of order 1%, i.e. subleading to the current vertex-reweighting systematic, but non-zero and asymmetric in z.

The more important consequence is the large *difference in the drop fraction* between single- and double-interaction productions (23% vs. 5%). When the double-interaction blend is applied (`run_showershape_double.sh`, `mix_weight` = 0.224 at 0 mrad and 0.079 at 1.5 mrad), the blended "no reco vertex" fraction becomes $0.776 \cdot 22.8\% + 0.224 \cdot 5.1\% \simeq 18.8\%$ (0 mrad) and $0.921 \cdot 22.8\% + 0.079 \cdot 7.9\% \simeq 21.4\%$ (1.5 mrad). This means the blended MC has a meaningfully lower "vertex-finder failure rate" than the nominal single-interaction MC, which (a) should be cross-checked against the observed data vertex-finding efficiency and (b) couples the MBD vertex efficiency systematic to the pile-up fraction systematic in a way not currently captured by the shower-shape double-interaction variation.

Recommendations

1. Document the MBD vertex efficiency ($\sim$77% in single-interaction MC, $\sim$95% in double) and the `vertex_weight` = 0 drop behaviour in the analysis note, next to the vertex reweighting description.
2. Cross-check the MBD vertex efficiency on data (e.g. fraction of MB-triggered events passing the same chain that fails the `MbdVertexMap`/`GlobalVertexMap::MBD` requirement) and compare to the blended-MC value above.
3. Consider a simple systematic: rerun the efficiency with `vertex_weight` = 1 fallback for out-of-range events and take the cross-section shift as the "MBD vertex efficiency modelling" uncertainty.

Files

- `efficiencytool/TruthVertexRecoCheck.C` — filler macro
- `efficiencytool/TruthVertexRecoCheckDraw.C` — drawing macro
- `results/truth_vertex_reco_check.root` — histograms and canvases
- `reports/figures/truth_vertex_reco_check.pdf` — shape plot
- `reports/figures/truth_vertex_reco_check_ratio.pdf` — ratio plot